

- **Weekly:** A new release is available every week to fix bugs and provide features to the community.

LTS and weekly releases are available in different flavours such as `.war` files (Jenkins is written in Java), native packages for the operating systems, installers and Docker containers.

The current LTS version is Jenkins 2.73.3. This version comes with a very useful option, called Deploy to Azure. Yes, we can deploy Jenkins to the Microsoft Azure public cloud within minutes. Of course, you need a Microsoft Azure subscription to utilise this option. Jenkins can be installed and used in Docker, FreeBSD, Gentoo, Mac OS X, OpenBSD, openSUSE, Red Hat/Fedora/CentOS, Ubuntu/Debian and Windows.

The features of Jenkins are:

- Support for SCM tools such as Git, Subversion, Star Team, CVS, AccuRev, etc.
- Extensible architecture using plugins: The plugins available are for Android development, iOS development, .NET development, Ruby development, library plugins, source code management, build tools, build triggers, build notifiers, build reports, UI plugins, authentication and user management, etc.
- It has the 'Pipelines as a Code' feature, which uses a domain-specific language (DSL) to create a pipeline to manage the application's lifecycle.
- The master agent architecture supports distributed builds.

To install Jenkins, the minimum hardware requirements are 256MB of RAM and 1GB of drive space. The recommended hardware configuration for a small team is 1GB+ of RAM and 50 GB+ of drive space. You need to have Java 8 - Java Runtime Environment (JRE) or a Java Development Kit (JDK).

The easiest way to run Jenkins is to download and run its latest stable WAR file version. Download the `jenkins.war` file, go to that directory and execute the following command:

```
java -jar jenkins.war.
```

Next, go to `http://localhost[IP address]:8080` and wait until the 'Unlock Jenkins' page appears. Follow the wizard instructions and install the plugins after providing proxy details (if you are configuring Jenkins behind a proxy).

Configuration

- To install plugins, go to **Jenkins Dashboard > Manage Jenkins > Manage Plugins**. Verify the updates as well as the available and the installed tabs. For the HTTP proxy configuration, go to the **Advanced** tab.
- To manually upload plugins, go to **Jenkins Dashboard > Manage Jenkins > Manage Plugins > Advanced > Upload Plugin**.
- To configure security, go to **Jenkins Dashboard > Manage Jenkins > Configure Global Security**. You can

configure authentication using Active Directory, Jenkins' own user database, and LDAP. You can also configure authorisation using Matrix-based security or the project-based Matrix authorisation strategy.

- To configure environment variables (such as `ANDROID_HOME`), tool locations, SonarQube servers, Jenkins location, Quality Gates - Sonarqube, e-mail notification, and so on, go to **Jenkins Dashboard > Manage Jenkins > Configure System**.
- To configure Git, JDK, Gradle, and so on, go to **Jenkins Dashboard > Manage Jenkins > Global Tool Configuration**.

Creating a pipeline for Android applications

We have the following prerequisites:

- Sample the Android application on GitHub, GitLab, SVN or file systems.
 - Download the Gradle installation package or configure it to install automatically from Jenkins Dashboard.
 - Download the Android SDK.
 - Install plugins in Jenkins such as the Gradle plugin, the Android Lint plugin, the Build Pipeline plugin, etc.
- Now, let's look at how to create a pipeline using the Build Pipeline plugin so we can achieve the following tasks:
- Perform code analysis for Android application code using Android Lint.
 - Create an APK file.

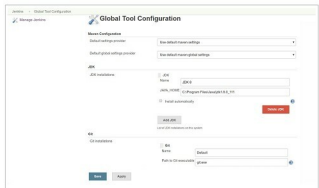


Figure 1: Global tool configuration

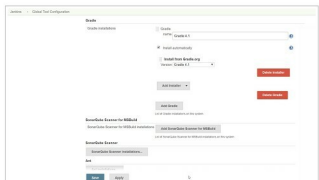


Figure 2: Gradle installation